

Mahesh Shantaram

(+91) 99801 29770 · ms@thecontrarian.in · Bengaluru, India

open data · interactive visualisation · statistical analysis · editorial technology

Summary

Over a two-decade career as a creative professional, I produced long-form photo essays for international audiences — commissioned by and featured in *The New York Times Magazine*, *TIME Magazine*, *Bloomberg Businessweek*, *BBC Culture*, *The Economist*, *Financial Times*, *Foreign Policy*, *Vanity Fair* to name a few. That background shaped how I think about the tension between evidence and narrative, fact and the visual, and the people who have to make sense of it all.

Since 2024, I've extended that practice into data — using Python, SQL, statistical tools, machine learning, and a modern web stack to build open datasets, reproducible pipelines, analytical tools, and immersive data stories. Recent work spans visualisation of Bengaluru's traffic and metro ridership patterns, live Indian election results, complex network relationships, digitally-signed content credentials and solar flare emissions from ISRO's Aditya-L1 observatory.

Skills

Skills used in actual open source projects available at github.com/thecont1

Data & Analysis [Python](#) · [SQL](#) · [pandas](#) · [NumPy](#) · [SciPy](#) · [Jupyter](#) · [statistical hypotheses testing](#) · [statsmodels](#) · [scikit-learn](#) · [DuckDB](#) · [data cleaning](#) · [web scraping](#) · [Selenium](#) · [BeautifulSoup](#) · [data validation](#) · [time-series analysis](#) · [PostgreSQL](#) · Visualisation & ProductDev [TypeScript](#) · [JavaScript](#) · [React](#) · [D3.js](#) · [Plotly](#) · [Cytoscape.js](#) · [Three.js](#) · [GSAP](#) · [Vite](#) · [Tailwind CSS](#) · [Recharts](#) · [Leaflet / mapping \(GIS\)](#) · [scrollytelling](#) · [accessible design](#) · Systems & Delivery [FastAPI](#) · [Rust](#) · [Topcoat \(custom Rust framework\)](#) · [Cloudflare Workers](#) · [Docker](#) · [GitHub Actions](#) · [Git](#) · [REST APIs](#) · [OpenAPI](#) · [WebMCP](#) · [E2E testing/Playwright](#) · [pytest](#) · [Fly.io](#) · [Vercel](#) · [Bun](#) · [OCR/document AI](#) · [LLM integration](#) · Domains [Urban mobility](#) · [public transport](#) · [civic data](#) · [content provenance](#) · [scientific datasets](#) · [data journalism](#) · [desktop applications](#) · [publishing tooling](#)

Experience

Data Scientist — Petasight, Inc. (formerly Babbage Insight)

Full-time · Bengaluru, India · Nov 2025 – May 2026

Petasight is a statistical analysis engine that turns raw enterprise-wide data into analyst-grade daily briefings.

- Petasight Insight Engine — Owned performance work that cut a single insight-generation run from >1 hr to ~7 min and per-run cost from >\$20 to ~\$2.50.
- Babbage Insight newsletter — Diagnosed and fixed a bug in the insight-scoring formula in the first week of joining. This caused the insight ranking to become explainable.
Stack: Python, FastAPI, PostgreSQL, AWS, Xiaomi MiMo 2.5 Pro

Visual Data Storyteller & Data Pipeline Plumber

Independent · Bengaluru, India · Jun 2024 – Present

Building open datasets, reproducible data pipelines, analytical tools, and interactive data stories on public-interest topics.

- [traffiCOracle](#) — End-to-end urban-mobility data platform for Bengaluru: automated collection of route-level travel-time and weather data, a statistical toolkit (anomaly detection, stationarity/variance testing, the R^3S^2 route-reliability score), and a zero-backend interactive dashboard visualising live vs. historical traffic conditions with uncertainty bands. *Python, FastAPI, pandas, Selenium, Docker, Cloudflare Workers, React, TypeScript, D3, Playwright.*
- [Namma Metro Ridership](#) — Longitudinal public dataset preserving Bengaluru Metro ridership figures (2+ years) not otherwise archived by the source; automated ETL with validation and anomaly checks; statistical investigation into fare-media reporting discrepancies; public dashboard with calendar-heatmap and travel-pattern views. *Python, pandas, Selenium, GitHub Actions, Rust, Topcoat, D3, Docker, Fly.io.*
- [India Votes Data](#) — Live Indian election results collection, verification, and analytics platform; concurrent rate-limited scraping of Election Commission sources into a structured database; live counting-day dashboard; OCR/vision-model-assisted cross-verification against official Form 20s. *Python, FastAPI, Pydantic, SQLite/PostgreSQL, Selenium, Tesseract OCR, Chart.js.*
- [Relex](#) — Interactive network-visualisation application built for IISc to explore relationships among faculty, research domains, projects, and platforms; reusable, multi-tenant, config-driven architecture with accessible 2D/3D and table views. *TypeScript, React, Cytoscape, Three.js, SheetJS.*
- [C2PA Viewer](#) — API-driven tool for verifying image provenance (C2PA credentials, EXIF/IPTC metadata), separating low-cost metadata reads from cryptographic verification. *Python, FastAPI, c2pa-python, Docker, Fly.io.*
- [SoLEXS Solar Flare Explorer](#) — Interactive notebook processing ISRO Aditya-L1 satellite data (FITS → Parquet), with configurable flare detection and light-curve visualisation. *Python, Astropy, pandas, Matplotlib.*
- [FlatWrite](#) — Markdown-native publishing tool with custom typography/pagination engines, HTML/PDF export, a public rendering API, and an MCP server for AI-agent integration. *TypeScript, Node.js, MCP, Paged.js, Vivliostyle, Cloudflare Workers.*
- [Quick2AVault](#) — macOS personal-finance app for GAAP-aware document management: deduplication, searchable Markdown conversion, AI-assisted extraction, local SQLite storage. *TypeScript, React, Glaze SDK, SQLite, Tailwind CSS.*
- [thecontrarian.in](#) — Personal platform unifying photography archive of 20 years, essays, code experiments, and data stories, with C2PA-verified images and DuckDB-backed analysis. *Astro, React, DuckDB, Cloudflare R2.*

Documentary Photographer

Independent · Bengaluru, India · Jun 2006 – Present

Photography, media, and creative professional with an international arts, journalism and publishing career.

- Produced long-form documentary projects on social and political themes in India, published and commissioned by world-renowned publications in the USA, Europe and Great Britain.
- Notable works include *Matrimania* (ft. *Financial Times*, *The Guardian*, the *BBC*) and *The African Portraits*, an award-winning series on racism in India (ft. *The Economist*).
- 2018 Alkazi Foundation *Photobook of the Year Award* — Winner
- 2015 British Journal of Photography — *Ones to Watch* (annual global talent list)

- 2011 Better Photography *Wedding Photographer of the Year (Portraiture)* — Winner
- Member Photographer — **Agence VU'**
 - Active member of the storied photography agency based in Paris*
 - Full-time · Paris, France (Remote) · *Sep 2017 – Apr 2020*
 - *Mémoire collective* — Long-form project commissioned by the International Federation of Human Rights (FIDH) in Guinea-Conakry on the occasion of the 60th anniversary of Guinea's Independence

Research Analyst — Corporate Executive Board (Gartner)

Full-time · Washington, D.C., United States · *2003 – 2005*

- Conducted primary research through interviews of C-suite executives to understand best-practices and find emerging patterns for industry.

Channel Editor — ZDNet India

Full-time · Mumbai, India · *2000 – 2003*

- Edited daily tech coverage; developed tech workflows for the editorial team.

Technology Journalist — CIOI.com

Full-time · Bangalore, India · *1999 – 2000*

- Researched and wrote technology articles for India's first tech news and features portal.

Education

Scaler — Data Science & Machine Learning (DSML) Certificate · *Jun 2024 – Jun 2026*

Ulster University (Belfast School of Art) — MFA Photography · *2021 – 2023*

- Featured in SOURCE Graduate Photography Online 2023
- Featured in Irish Arts Review, "New Generation Gallery / Class of 2023"
- 2022 Alice Berger Hammerschlag Award, Winner, MFA Project

Spéos Institute, Paris Diploma in Photography, *avec mention* · *2006*

St Joseph's College of Arts & Science B.Sc. (Physics, Mathematics, Computers) · *1998*

Mallya Aditi International School ISC · *1995*

St. Germain High School ICSE · *1993*

Press

[BBC Culture](#): "Matrimania: The changing face of India's weddings" (16 Sep 2019)

[Financial Times](#): "Why weddings are a perfect window into modern India" (15 Aug 2018)

[Foreign Policy](#): "Out of India – A wave of brutal violence against visiting college students is forcing the country to examine its racism problem" (24 July 2017)

[The Economist](#): "Intimate portraits of Africans in India" (19 May 2017)